

ZUGSPITZE, Germany

WMO: 109610

Lat: 47.4208N

Lon: 10.9847E

Elev: 9712

StdP: 10.22

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.3	-5.8	-24.5	2.0	6.0	-19.3	2.7	10.0	56.4	13.4	50.6	15.6	13.5	0	1.095

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	51.8	44.1	49.2	42.1	47.0	40.8	46.0	49.3	44.1	47.1	42.5	45.5	8.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
44.6	62.7	46.7	42.7	58.4	45.2	41.0	54.5	43.3	21.3	49.7	20.1	47.2	19.2	45.7	52.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 47.9	40.5	33.6	DB	-14.4	57.7	6.4	2.3	-19.1	59.4	-22.8	60.7	-26.5	62.0	-31.1	63.7	(4)
(5)			WB	-14.6	50.0	6.5	1.8	-19.3	51.3	-23.1	52.4	-26.7	53.4	-31.4	54.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.5	13.7	11.6	14.3	18.6	27.0	32.3	38.4	37.9	34.0	28.8	20.2	16.3		
	DBStd	12.18	9.12	9.48	8.30	7.07	6.91	6.92	6.54	6.70	6.94	7.63	8.42	8.58		
	HDD50	9309	1127	1075	1108	941	713	531	361	376	481	657	895	1044		
	HDD65	14781	1592	1495	1573	1391	1178	981	824	840	931	1122	1345	1509		
	CDD50	3	0	0	0	0	0	0	2	1	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	15.2	19.1	16.5	16.3	15.0	12.4	12.9	12.3	12.5	14.3	15.7	17.1	18.3		
	PrecAvg	79.1	7.0	6.0	7.5	7.4	7.0	7.2	7.3	7.0	5.2	4.6	6.3	7.0		
(15) Precipitation	PrecMax	107.2	17.3	16.1	17.8	21.8	13.7	13.7	15.0	14.4	13.6	11.9	14.4	18.8		
	PrecMin	55.7	0.3	1.2	1.1	0.6	0.8	3.7	3.3	2.4	1.1	0.2	0.0	0.6		
	PrecStd	12.2	4.1	3.3	3.6	4.0	3.2	2.3	2.4	2.4	2.7	3.0	3.3	4.0		
	PrecStd	12.2	4.1	3.3	3.6	4.0	3.2	2.3	2.4	2.4	2.7	3.0	3.3	4.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.6	35.2	33.3	37.8	44.3	51.1	56.6	55.7	50.4	46.8	39.6	34.8		
		MCWB	24.3	23.8	26.2	29.4	37.4	43.1	44.6	45.4	40.0	35.7	26.8	26.1		
	2%	DB	29.8	30.8	29.9	33.6	40.9	46.9	52.4	51.5	47.4	43.2	36.8	32.1		
		MCWB	21.6	22.8	24.4	27.7	35.8	41.2	44.3	44.2	38.6	33.7	25.8	23.5		
	5%	DB	27.4	27.2	27.7	30.4	37.9	44.2	49.3	49.0	45.0	40.7	34.1	29.6		
		MCWB	20.4	21.3	23.2	26.1	34.0	39.8	42.9	42.3	37.6	32.5	25.2	23.3		
	10%	DB	24.6	24.3	25.2	27.9	35.2	41.7	46.9	46.5	42.8	38.7	31.1	27.2		
		MCWB	19.8	20.6	21.9	25.0	32.4	38.8	41.9	41.0	36.4	31.4	24.0	21.9		
	0.4%	WB	27.1	26.1	27.8	31.6	40.5	46.3	49.0	48.9	43.0	38.5	29.8	29.9		
		MCDB	29.8	31.0	30.7	35.1	42.9	48.1	51.8	51.7	47.0	44.0	32.3	31.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	24.4	24.2	25.9	28.9	37.2	43.1	46.3	46.0	41.1	36.0	28.5	26.7		
		MCDB	26.5	27.9	28.6	32.0	39.7	45.0	49.6	49.6	45.0	39.8	31.8	28.9		
	5%	WB	22.1	22.6	24.2	27.3	34.8	41.0	44.4	44.0	39.5	34.6	27.0	24.9		
		MCDB	25.1	26.7	26.6	29.4	36.7	43.3	47.4	47.3	43.0	38.4	30.7	27.3		
	10%	WB	19.9	20.9	22.3	25.7	32.9	38.9	42.8	42.3	38.1	33.1	25.6	22.8		
		MCDB	23.6	24.0	24.7	27.4	34.7	41.2	46.0	45.2	41.1	36.8	29.9	25.9		
	5% DB	MDBR	9.1	9.1	9.1	9.2	8.7	8.3	8.9	8.7	8.4	8.4	8.5	8.9		
		MCDBR	10.2	10.3	9.8	11.0	10.3	10.7	11.5	11.2	9.7	9.2	8.6	9.4		
		MCWBR	7.3	6.9	8.2	9.2	8.8	10.2	11.6	11.5	8.5	6.8	5.6	7.6		
		MCWBR	10.1	9.6	9.7	9.8	9.0	9.8	10.5	10.5	8.8	8.4	8.5	9.5		
(35) Mean Daily Temperature Range	5% WB	MCWBR	8.8	8.1	9.1	8.9	8.6	10.1	11.0	10.9	8.3	7.2	7.4	9.1		
		MCWBR	8.8	8.1	9.1	8.9	8.6	10.1	11.0	10.9	8.3	7.2	7.4	9.1		
	Clear-Sky Solar Irradiance	taub	0.193	0.204	0.225	0.247	0.258	0.265	0.265	0.263	0.247	0.230	0.211	0.192		
		taud	2.572	2.555	2.595	2.625	2.643	2.652	2.642	2.667	2.708	2.734	2.690	2.598		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	307	321	326	324	320	317	315	314	313	308	298	296		
		Edh at Noon	17	22	25	27	28	28	28	26	23	19	16	15		
(44) All-Sky Solar Radiation	RadStd	RadAvg	425	720	1059	1356	1519	1706	1693	1502	1117	770	445	351		
		RadStd	51	90	118	179	175	173	162	163	136	87	67	49		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (93 neighbors)	N/A	N/A	N/A	N/A	+0.79	+0.76	N/A	N/A	+101	+23

Nomenclature: See separate page